

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q21-1643

West -Station 138 kV  
November 28, 2021

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1643. The Interconnection Customer proposes a 75.4 MW solar photovoltaic facility interconnecting at the West -Station 138 kV substation in the Maryland area. The request was received on January 02, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 614 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,329,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                   | Contingency                                       | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------|---------------------------------------------------|-------------|--------------|----------|
| SANDBURG to JARRATT 138.0 kV         | VIENNA OPERATIONS to EASTON 2 SUBSTATION 345.0 kV | 150.2       | 542          | 27.4     |
| SANDBURG to SHERWOOD 138.0 kV L      | VIENNA OPERATIONS to EASTON 2 345.0 kV            | 148.9       | 554          | 27.4     |
| HITCHLAND to SHERWOOD 138.0 kV L     | VIENNA OPERATIONS to EASTON 2 345.0 kV            | 148.2       | 561          | 27.4     |
| NORTH LAKE to KIWI 138.0 kV Line     | VIENNA OPERATIONS to EASTON 2 345.0 kV            | 147.4       | 528          | 23.7     |
| ESS LEWES to NEWMARK 138.0 kV        | EASTON2 to VIENNA OPERATIO 345.0 kV Line          | 146.8       | 427          | 20.9     |
| NORTH LAKE to HITCHLAND 138.0 kV     | EASTON2 SUBSTATION to VIENNA OPERATIONS 345.0 kV  | 145.3       | 591          | 27.4     |
| CHEMOLITE to LAMMRS 138.0 kV         | VIENNA OPERATIONS to EASTON 2 345.0 kV            | 137.9       | 318          | 16.6     |
| WYE MILLS to HOLSTON DAM 138.0 kV    | CADY to EASTON 2 345.0 kV                         | 131.2       | 231          | 5.8      |
| PARK AVE to HILLSBORO 138.0 kV       | CADY to EASTON 2 345.0 kV                         | 130.8       | 760          | 25.0     |
| W -STATION to ST BONIFACIUS 138.0 kV | SOUTHTOWN to VIENNA OPERATIONS 345.0 kV           | 129.5       | 1,362        | 72.0     |
| CHEMOLITE to RANDALL COUNTY 138.0 kV | VIENNA OPERATIONS to EASTON 2 345.0 kV            | 129.0       | 330          | 16.6     |
| ROBINSONVILLE to ANGOA 138.0 kV Line | QUINN to BULVERDE 138.0 kV                        | 127.6       | 180          | 5.4      |

| Monitored Facility                            | Contingency                                        | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------|----------------------------------------------------|-------------|--------------|----------|
| PARK AVE to HARBOR 138.0 kV                   | CADY to EASTON 2 345.0 kV                          | 124.8       | 410          | 9.5      |
| GRASONVILLE to MACON 138.0 kV line            | EASTON 2 to CADY 345.0 kV                          | 123.7       | 433          | 11.3     |
| CADY 345/138 kV TX                            | DODD RD to CADY 345.0 kV                           | 123.4       | 823          | 22.9     |
| CARANCAHUA to HOLSTON DAM 138.0 kV Line       | CADY to EASTON 2 345.0 kV                          | 123.3       | 235          | 5.5      |
| HARBOR to HOLSTON DAM 138.0 kV                | EASTON 2 to CADY 345.0 kV L                        | 123.1       | 432          | 9.5      |
| OAK HILL to LOUISA 138.0 kV line              | CADY to DODD ROD 345.0 kV                          | 122.6       | 324          | 9.6      |
| LOST HILLS to OAK HILL SUBSTATION 138.0 kV    | CADY to DODD ROAD 345.0 kV                         | 122.3       | 466          | 14.5     |
| CADY to LOUISA 138.0 kV                       | CADY to DODD ROAD 345.0 kV                         | 121.3       | 338          | 9.6      |
| MESQUITE SUBSTATION to ARROYO TAP 138.0 kV L  | VIENNA OPERATIONS SUBSTATION to SOUTHTOWN 345.0 kV | 119.9       | 1,151        | 27.8     |
| OCEAN CITY to REIDLAND 138.0 kV               | VIENNA OPERATIONS to SOUTHTWN SUBSTATION 345.0 kV  | 119.3       | 1,171        | 27.8     |
| ARROYO TAP to REIDLAND 138.0 kV               | SOUTHTOWN SUBSTATION to VIENNA OPERATIONS 345.0 kV | 119.3       | 1,171        | 27.8     |
| CANDELARIA ARENAS 1718 to OCEAN CITY 138.0 kV | VIENNA OPERATIONS to SOUTHTOWN 345.0 kV            | 118.9       | 1,187        | 27.8     |
| CADY to KEY LARGO 138.0 kV                    | CADY to DODD ROAD 345.0 kV                         | 116.1       | 214          | 5.5      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------|---------------------------------------------|---------------|-----------------------|
| SANDBURG to JARRATT 138.0 kV      | Rebuild line with higher-capacity conductor | \$9,483,000   | 2% (\$235,000)        |
| SANDBURG to SHERWOOD 138.0 kV L   | Rebuild line with higher-capacity conductor | \$3,424,000   | 2% (\$85,000)         |
| SHERWOOD to HITCHLAND 138.0 kV    | Rebuild line with higher-capacity conductor | \$6,786,000   | 2% (\$168,000)        |
| NORTH LAKE to KIWI 138.0 kV Line  | Rebuild line with higher-capacity conductor | \$5,309,000   | 2% (\$126,000)        |
| ESS LEWES to NEWMARK 138.0 kV     | Rebuild line with higher-capacity conductor | \$11,819,000  | 2% (\$246,000)        |
| N LKE to HITCHLAND 138.0 kV L     | Rebuild line with higher-capacity conductor | \$6,481,000   | 2% (\$161,000)        |
| CHEMOLITE to LAMMERS 138.0 kV     | Rebuild line with higher-capacity conductor | \$7,878,000   | 3% (\$208,000)        |
| WYE MILLS to HOLSTON DAM 138.0 kV | Rebuild line with higher-capacity conductor | \$6,552,000   | 2% (\$125,000)        |

| Facility                                             | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| PARK AVE to HILLSBORO 138.0 kV                       | Rebuild line with higher-capacity conductor | \$8,640,000   | 2% (\$186,000)        |
| ST BONIFACIUS to W -STATION SUBSTATION 138.0 kV line | Rebuild line with higher-capacity conductor | \$6,451,000   | 3% (\$200,000)        |
| CHEMOLITE to RANDALL COUNTY 138.0 kV                 | Rebuild line with higher-capacity conductor | \$6,496,000   | 3% (\$171,000)        |
| ANGOLA to ROBINSONVILLE 138.0 kV L                   | Rebuild line with higher-capacity conductor | \$5,567,000   | 2% (\$112,000)        |
| PARK AVE to HARBOR 138.0 kV                          | Rebuild line with higher-capacity conductor | \$18,831,000  | 2% (\$295,000)        |
| MACON to GRASONVILLE 138.0 kV line                   | Rebuild line with higher-capacity conductor | \$2,434,000   | 2% (\$45,000)         |
| CADY 345/138 kV TX                                   | Replace transformer with higher-rated unit  | \$13,336,000  | 2% (\$253,000)        |
| CARANCAHUA to HOLSTON DAM 138.0 kV line              | Rebuild line with higher-capacity conductor | \$13,572,000  | 2% (\$259,000)        |
| HARBOR to HOLSTON DAM 138.0 kV line                  | Rebuild line with higher-capacity conductor | \$22,769,000  | 2% (\$357,000)        |
| OAK HILL to LOUISA 138.0 kV line                     | Rebuild line with higher-capacity conductor | \$2,962,000   | 2% (\$58,000)         |
| LOST HILLS to OAK HILL SUBSTATION 138.0 kV           | Rebuild line with higher-capacity conductor | \$29,265,000  | 2% (\$574,000)        |
| LOUISA to CADY 138.0 kV                              | Rebuild line with higher-capacity conductor | \$4,012,000   | 2% (\$79,000)         |
| MESQUITE SUBSTATION to ARROYO TAP 138.0 kV L         | Rebuild line with higher-capacity conductor | \$1,736,000   | 2% (\$28,000)         |
| OCEAN CITY to REIDLAND 138.0 kV                      | Rebuild line with higher-capacity conductor | \$3,743,000   | 2% (\$60,000)         |
| ARROYO TAP to REIDLAND 138.0 kV                      | Rebuild line with higher-capacity conductor | \$1,313,000   | 2% (\$21,000)         |
| CANDELARIA ARENAS 1718 to OCEAN CITY 138.0 kV        | Rebuild line with higher-capacity conductor | \$2,280,000   | 2% (\$37,000)         |
| CADY to KEY LARGO 138.0 kV L                         | Rebuild line with higher-capacity conductor | \$11,992,000  | 2% (\$240,000)        |
|                                                      | Total allocated to project                  |               | \$4,329,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.